Natya Vidhan Biswas

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Experience

- **MIT Media Lab** — Cambridge, USA (Remote)
  Product Engineer — *Aug 2026 – Present*
  - **Machine Learning Workflows**: Build and maintain **AI/ML pipelines** spanning data preprocessing, model integration, and production inference.
  - **Full-Stack Development**: Own **back-end services** and **front-end interfaces** end-to-end for product features connecting ML systems to users.
- **University of Melbourne** — Melbourne, Australia (Remote)
  Machine Learning Engineer (Contract) — *Dec 2023*
  - **Multimodal Pipeline**: Built an end-to-end **multimodal Twitch analytics pipeline** processing **13+ TB** of video, audio, and live chat across **40** streamers to produce minute-level **sentiment, toxicity, and emotion** signals.
  - **Computer Vision**: Implemented **GPU-accelerated face identity filtering** using **InsightFace** embeddings and cosine similarity to robustly isolate streamer face frames across VOD segments; classified emotions per-chunk with **DeepFace**.
  - **NLP & Audio Analysis**: Integrated **faster-whisper** for speech transcription and engineered audio features (pitch, RMS loudness, bitrate). Built a multi-model scoring layer combining **GoEmotions**, **Detoxify**, and **HateSonar** for chat and transcript analysis, achieving ∼**92%** accuracy on emotion classification.
  - **Tooling & Infrastructure**: Productionized the system with a **CUDA-enabled Docker** environment; built a **Flask-based human-in-the-loop face embedding builder** to bootstrap per-streamer identity recognition for the CV pipeline.
- **Facts-n-Data** — Remote
  Software Development Intern — *Mar 2024 – Apr 2025*
  - **E-Commerce Automation**: Built and maintained **30+** data scraping and automation scripts across **Amazon**, **FirstCry**, and **Myntra**, collecting data on **10,000+** products to enable large-scale analysis.
  - **Analytics & Monitoring**: Developed scripts for **BSR tracking**, **search term analysis**, and **real-time stock monitoring** to drive data-informed business decisions.
  - **Internal Tooling**: Refactored legacy codebases for improved efficiency and built internal tools including **user authentication** systems and **employee management dashboards**.
- **BLITZ, The Computer Science Society** — Keshav Mahavidyalaya, Delhi University
  Secretary — *Aug 2025 – Aug 2026*
  - **Web Development**: Designed and developed the **official society website** for both annual cycles, reaching **5,000+** students and serving as the primary digital presence for events and announcements.
  - **Volunteer Management**: During flagship events, **managed all volunteers** on-ground, assigning tasks and ensuring smooth execution across all event verticals.

Projects

- **YABBE** – Self-Hosted Browser-Based Video Editor — GitHub
  - **Full-Stack Architecture**: Self-hosted **FFmpeg**-powered editor with a **React/TypeScript** frontend and **Python FastAPI** backend, shipped as a multi-arch **Docker** image.
  - **AI Roto Brush & Tracking**: Integrated Meta's **EdgeTAM** (**ONNX Runtime**) for CPU-only object segmentation, plus sub-pixel **motion tracking** and a custom **chroma keyer** that beats FFmpeg's built-in keyer on edge accuracy.
  - **Real-Time Preview**: Built a lazy, cached **HLS preview** pipeline and dependency-ordered pre-rendering for nested, precomp-style sequences.
  - **Additional Features**: Also added a **keyframe graph editor**, **74 transitions**, **nested/precomp sequences**, **freeze frame**, **markers**, and **shape layers**.
- **Custom ESP32 Drone** – Full-Stack Hardware Build — Personal Project
  - **Flight Firmware**: Flashed and configured **ESP-FC** flight-controller firmware on **ESP32**; reverse-engineered the FlySky FS-CT6B transmitter's USB protocol to read RC channel data directly.

- **Signal Pipeline**: Converted **PWM to PPM** via an Arduino Nano intermediary, then migrated to **RMT-based** RC channel reading on the ESP32 for lower-latency control input.
- **Flight Stability**: Implemented a **complementary filter** for sensor fusion, fixed yaw-mixing sign errors, and debugged ISR race conditions and Wi-Fi radio interference affecting in-flight stability.
- **Live Tuning**: Enabled real-time **PID gain tuning** via AUX-channel knobs on the transmitter for on-the-fly flight characteristic adjustments.

- **INT8 Quantized Inference Engine** – On-Device ML in C Blog | GitHub
  - **From-Scratch Training Pipeline**: Implemented a full **MLP training pipeline** (forward/backward pass, **Adam optimizer**) in pure **NumPy** with zero ML-framework dependencies, training a **668-parameter**, **30-16-8-4** network to **97%** test accuracy on IMU-derived gesture features.
  - **INT8 Quantization**: Designed a **symmetric per-tensor INT8** quantization scheme with **INT32** bias accumulators and data-calibrated inter-layer requantization, compressing the model to a **1.1 KB** binary with under **2%** accuracy loss.
  - **Header-Only C Runtime**: Built a dependency-free, **header-only C inference engine** (˜250 LOC, no TFLite Micro/Edge Impulse) that loads the quantized binary from **SPIFFS** and classifies gestures from live **MPU6050** IMU data in real time on an **ESP32**; model-agnostic for any dense-only MLP, and being extended toward **sensorless torque estimation** for robotic actuators.

- **Planq** – Exam Preparation Platform Website | GitHub
  - **Question Bank**: Curated a database of **47,000+** previous year questions spanning **6 All-India entrance exams**, serving **240+** registered students with **500+** concurrent live sessions at peak load.
  - **Smart Scheduling**: Implemented an **Anki-like spaced repetition** algorithm that schedules topics based on individual student performance and preparation analytics.

- **Donut Shop** – Home Server Infrastructure
  - **Cluster Setup**: Built a **2-node Linux cluster** in a **master-worker** configuration orchestrated with **k3s** and **Longhorn** for persistent storage.
  - **Services**: Deployed and managed **NAS**, media streaming (**Plex/Jellyfin**), **VPN**, and various containerized services.

Skills

- **Languages**: Python, C, C++, JavaScript, SQL
- **Embedded & Hardware**: ESP32 / Arduino, FreeRTOS, RC & flight-control firmware, PWM/PPM signal processing, sensor fusion, motor control (BLDC, SimpleFOC, PID tuning)
- **Machine Learning**: INT8 quantization, on-device inference, faster-whisper, InsightFace, DeepFace
- **Databases**: MongoDB, Supabase, SQLite
- **Frameworks**: Flask, Next.js, Tailwind CSS, Selenium, Pygame, DearPyGui
- **DevOps**: Docker, k3s, Linux, Git

Education

- **Keshav Mahavidyalaya, Delhi University** Pitampura, Delhi
  Bachelor of Science in Computer Science; CGPA: 6.88 (Till 3rd Semester) *Aug 2024 – May 2028*

- **PM Shri Kendriya Vidhyalaya** Vigyan Vihar, Delhi
  10+2, PCM; 88%/73% *April 2012 – March 2024*

Achievements

- **Pariksha Pe Charcha 2024**: Presented **Anisha**, an AI-powered therapist with real-time emotion detection, directly to the Hon'ble Prime Minister of India.
- **Brainwave 2.0 Hackathon (2026)**: Secured **3rd place in the Hardware track** and **8th overall** as Finalists at the 36-hour hackathon by Dept. of IT, DTU, in association with AIMS-DTU.
- **Smart India Hackathon 2025**: Secured **1st Runner-Up** in the Internal Round at Keshav Mahavidyalaya, Delhi University.